جامعة المرقب
كلية تقنية المعلومات

Object Oriented Programming

د.حميـــــــدة المشرقي

Email: h.a.almashregi@elmergib.edu.ly

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المحاضرة الاولى

The Outline

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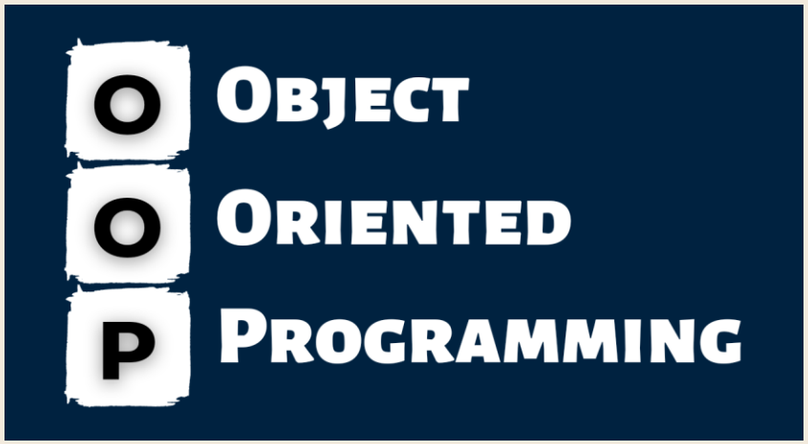
05 **Implementation class and its objects using C++**

Definition of OOP



- **Object oriented programming** : is a programming style that is intended to make thinking about programming closer to thinking about **the real world .**

- **The main idea**: Break down the program into independent units (objects), where each object represents something from the real world.

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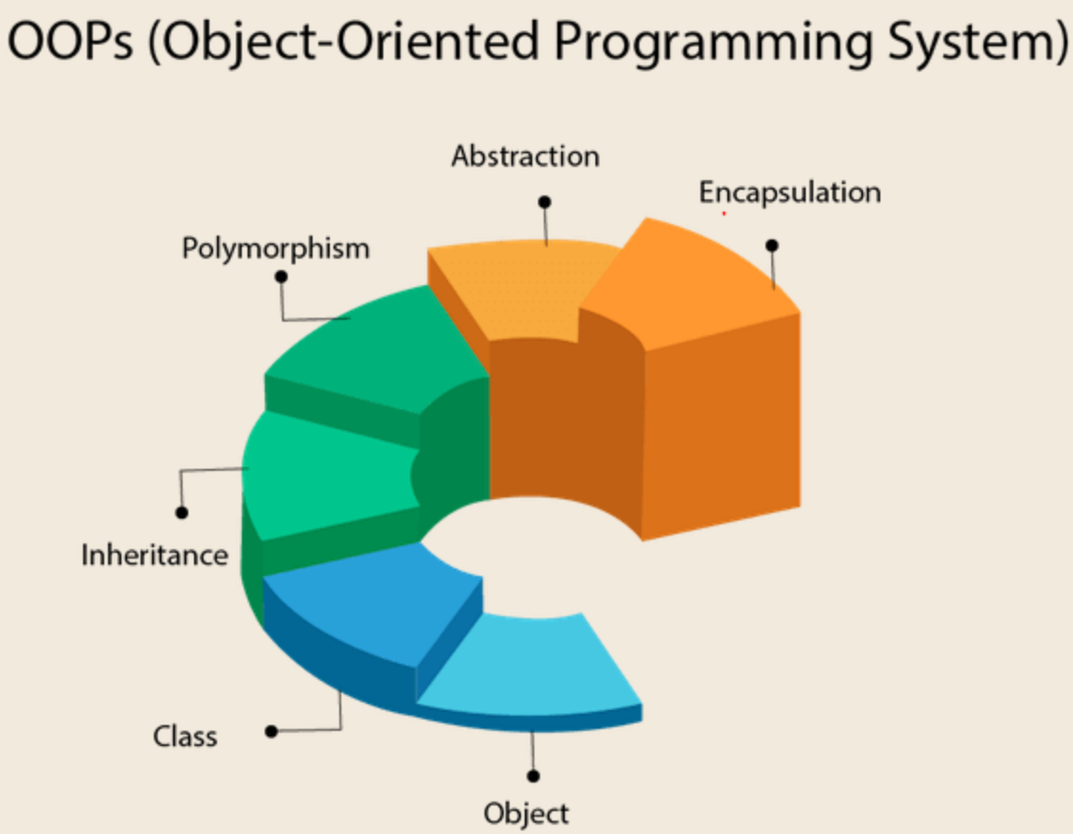
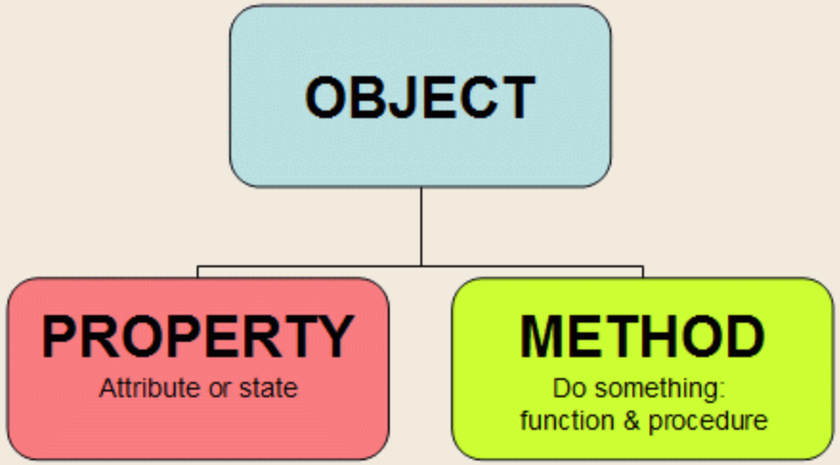
OOP Concepts



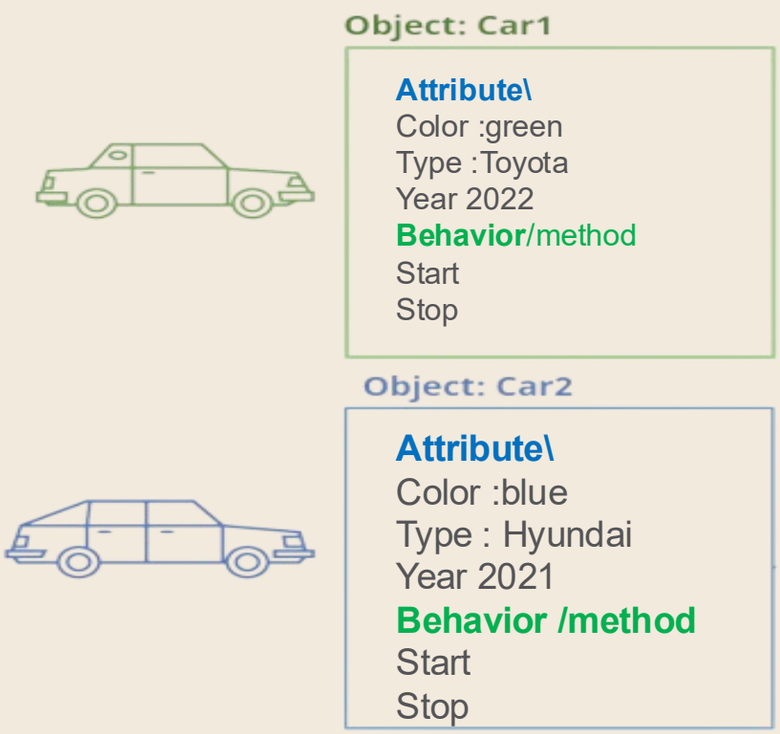
Object

- In programming objects are independent units and each has its own **identify** .just as objects in the real world do.
- A car is an object. So a person is an object , phone as well. each has own **identify.**



Object Attributes

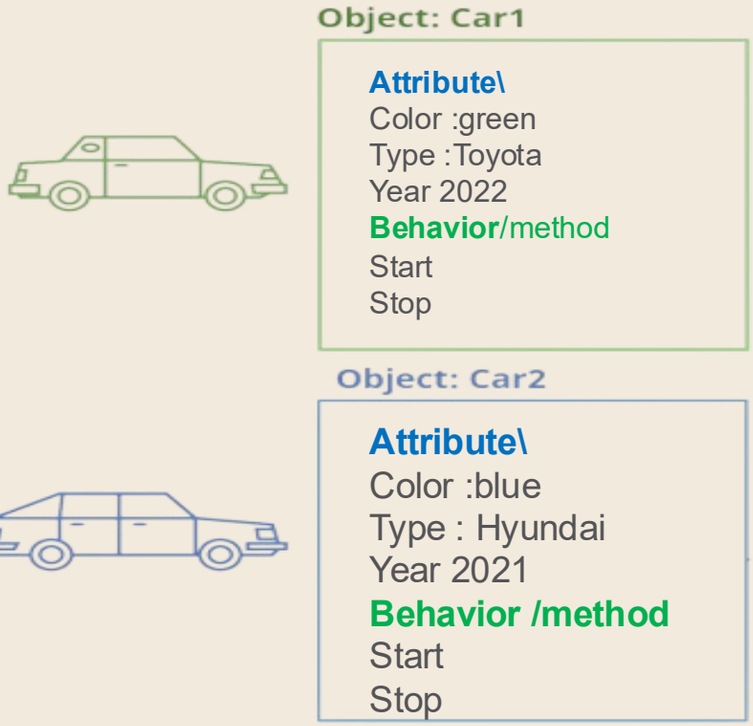
- Objects have characteristics called **attributes** that describe them.
- **Attributes (properties)** :describe the current **state** of an object.
- Example: A car can be blue or green, and can have multiple attributes like type and year as show in following image:



Object Methods

In the real world, each object has **behaviors** called **methods.**

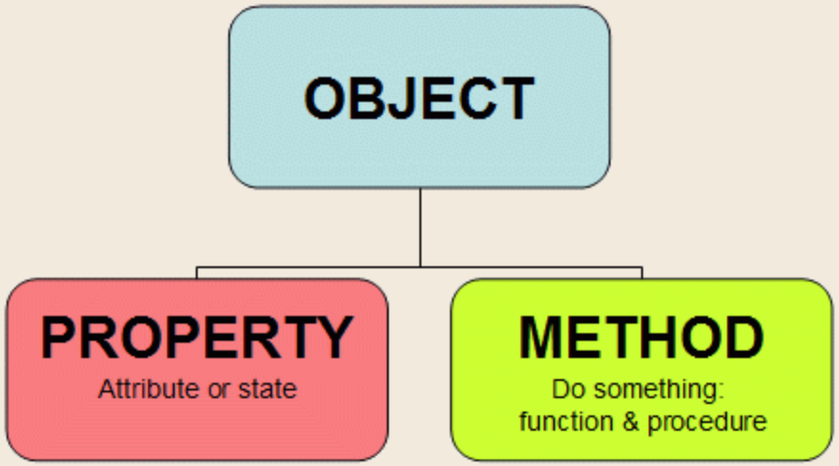
- **Methods** are the actions that objects can do.
- Example: A car can move, start, and stop - these are its methods.



Object

- So the following **three dimensions** describe any object in object oriented programming

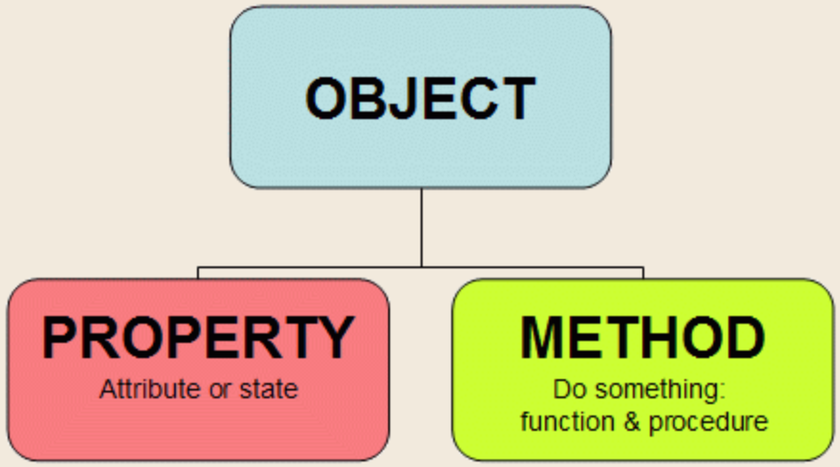
1. **Identity** - What type of object is it? the unique name given to an object by the user
2. **Attributes** - What characteristics does it have?
3. **Methods** - What actions can it do ?

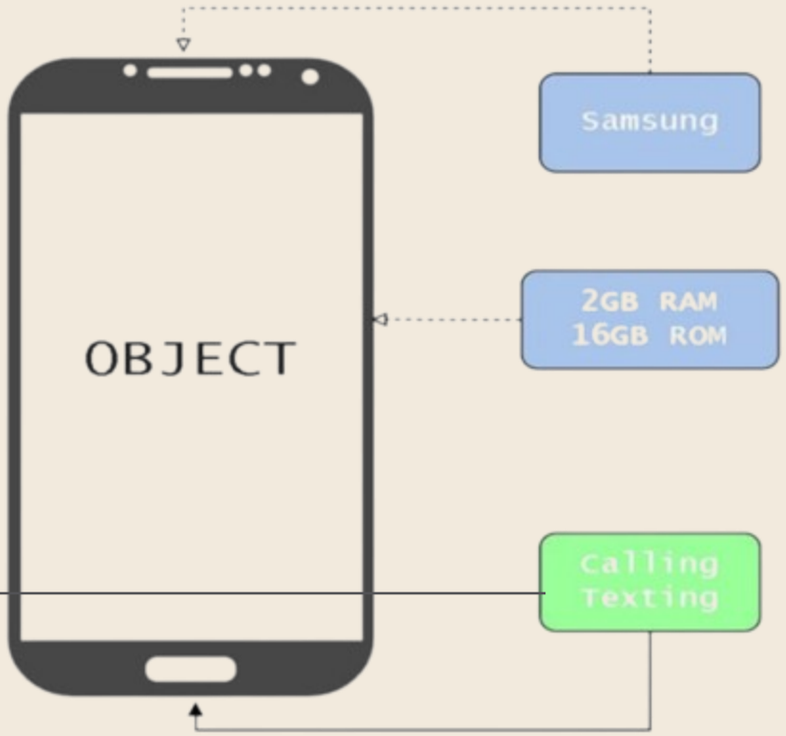


Object

According to the three dimensions of objects, explain the(Identity, Attributes, and Methods) for each object below:

Identify (object): Phone or any name
Attributes : name (Samsung)
Memory : 2GB RAM ,16GB ROM
Methods : calling ,texting





Object

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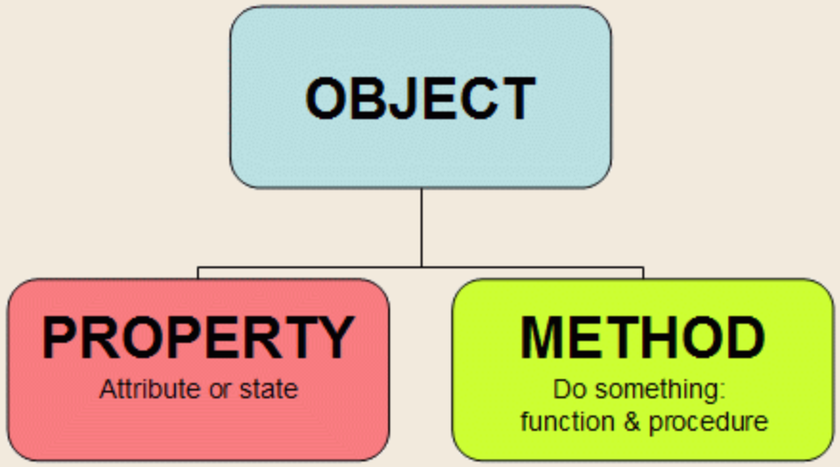
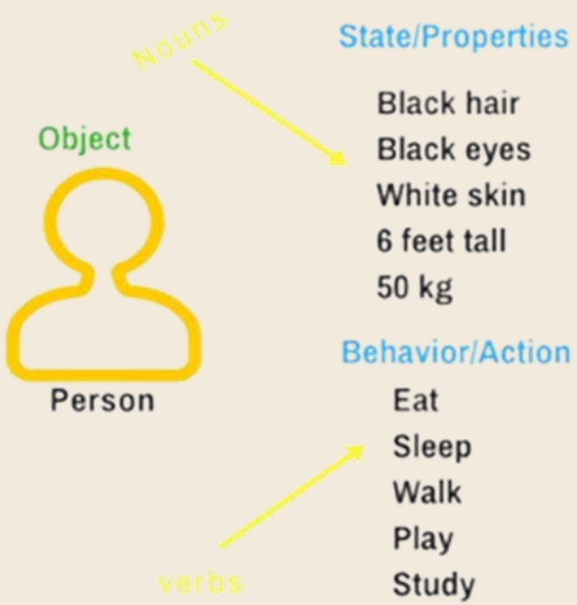
According to the three dimensions of objects, identify the(Identity, Attributes, and Methods) for each object below

Identify (object): person or any unique name
Attributes : black hair , black eyes white skin 6 feet tall ,50 kg (noun)
Methods : eat , sleep, walk , play , study (verb)





Class

- Definition: A blueprint or template that defines the specifications for an object. For example

  Class Car (The Blueprint) :
  - defines what every car should have
  - Contains common specifications for all cars

  Objects: Audi, Nissan, Volvo (Actual Cars)
  - Created from the Car class blueprint
  - Each is a specific instance of a car

Class

- A class is a **group of objects** that **share common properties and behavior**.

Figure A :All of three objects share the same structure (attributes and behaviors) defined by the Man class

Figure B :The object car attributes appears to be doesn't relate to the Man class.

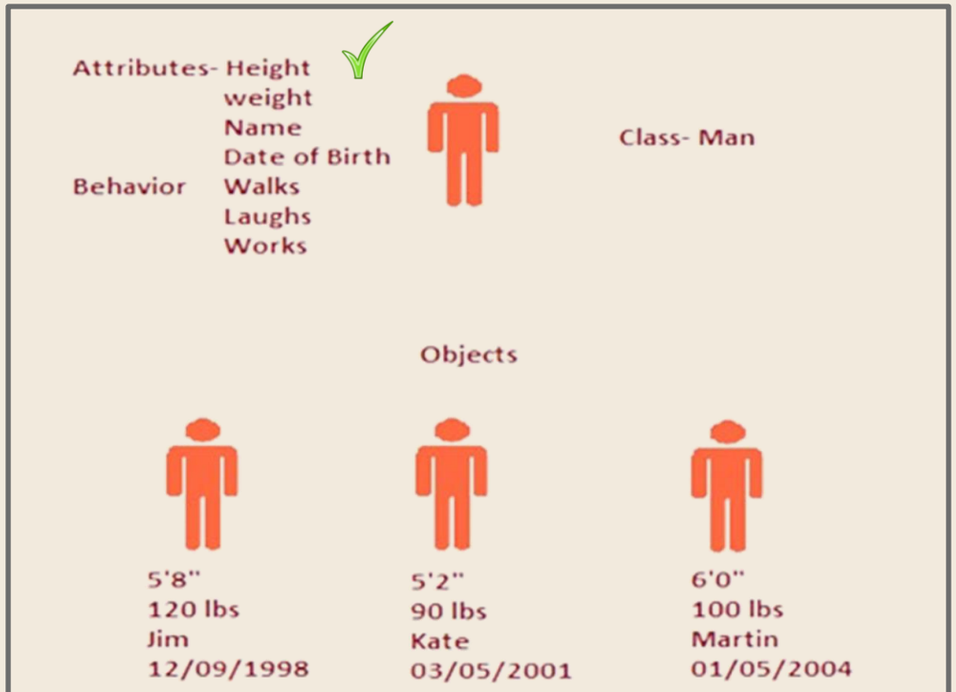


Figure A

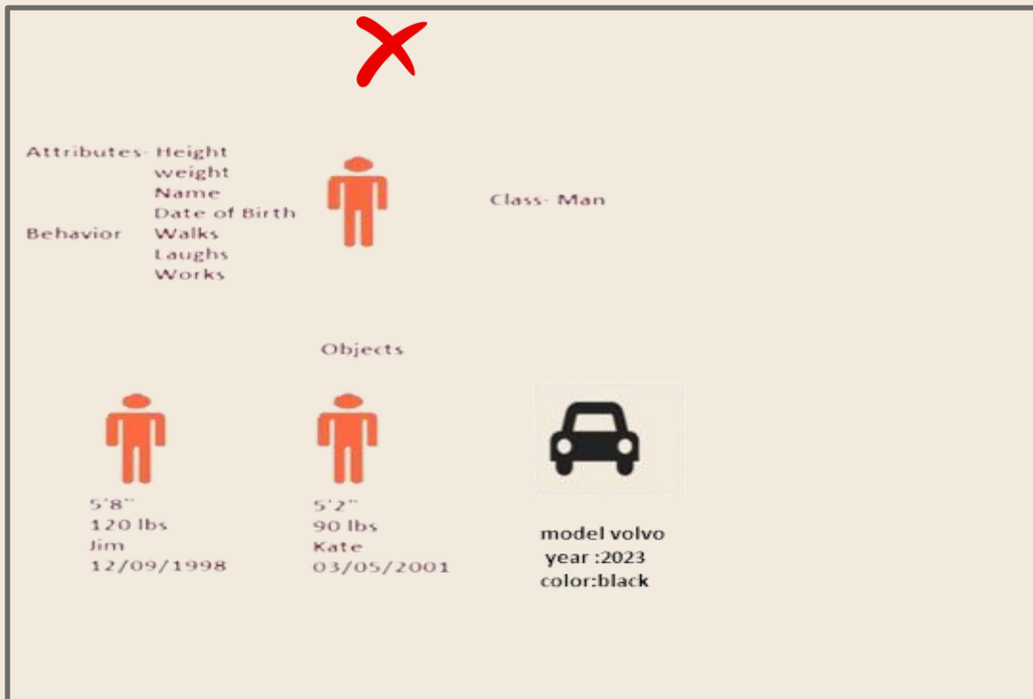


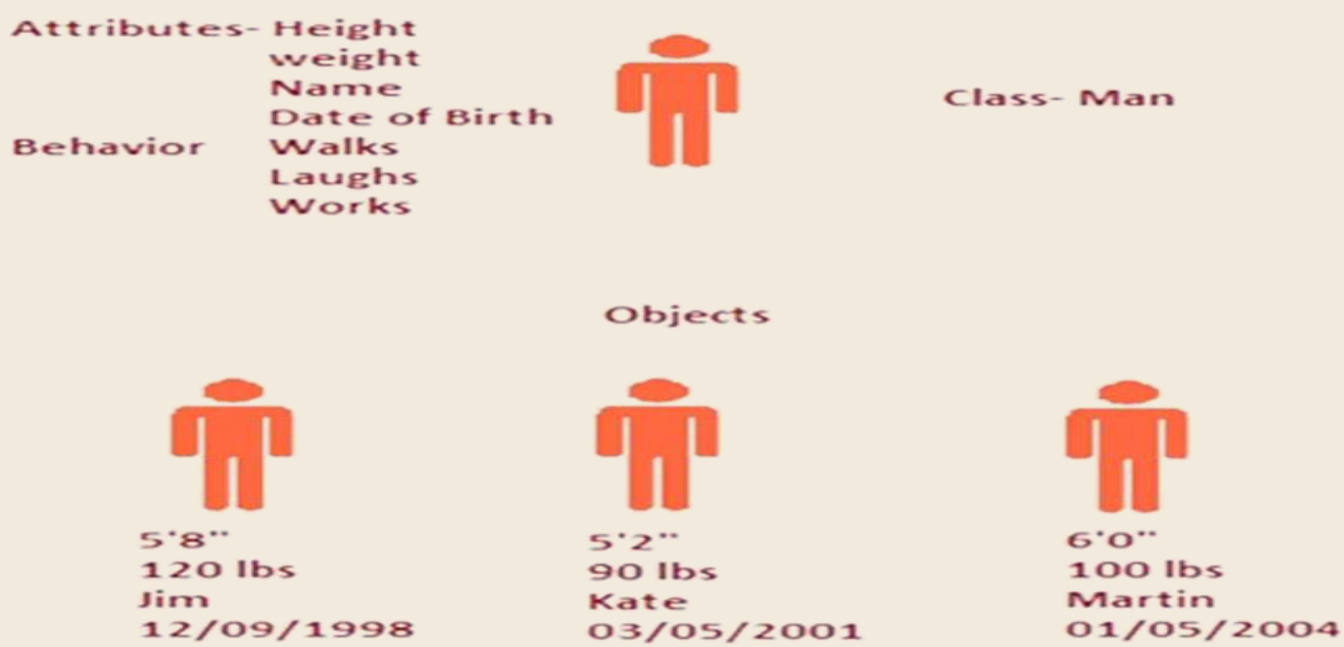
Figure B

Class

- **Objects are variables of class :**

From following figure : for all objects same attributes (height, weight , name ,data of birth) . Values of those attributes - different for each object :

- Object 1: 5.8ft ,120 lbs, Jim , 12/09/1998
- Object 2: 5.2ft ,90, lbs, kate, 08/05/2001
- Object 3: 6.0ft ,100, lbs,martin , 01/05/2004



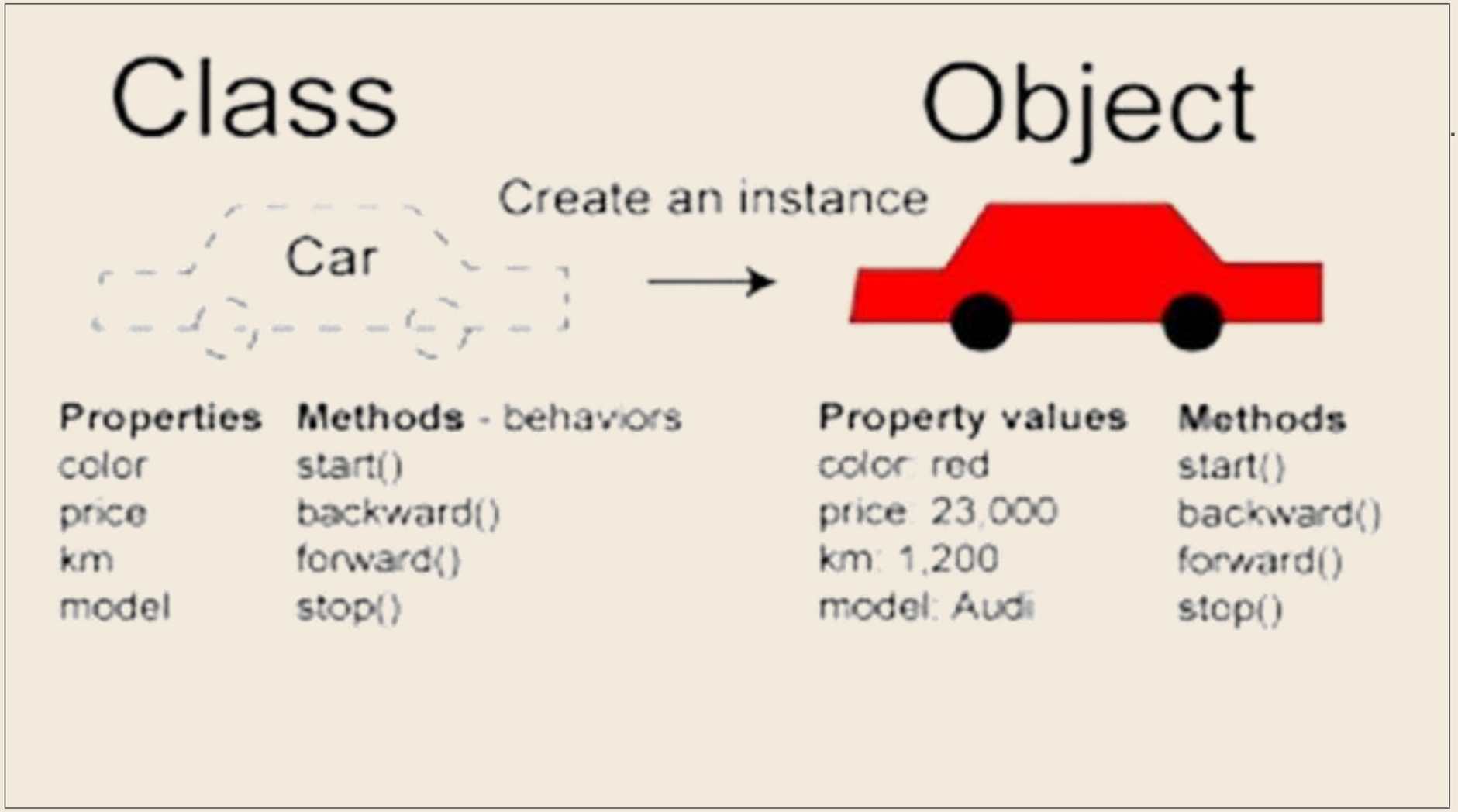
Implementing Class and Objects with C++

Now that we understand classes and objects, we will implement them in programming using C++. Before we start, let's mention the general steps for implementation:

1. Declare Class
2. Define Attributes
3. Define Methods
4. Create Objects
5. Set Property Values
6. Call Methods

Implementing Class and Objects with C++

Following the example of a Car class with properties and methods, we will now implement this class and object in C++ using previous steps



Step 1: Declare Classe Class

Declare class with keyword class then the name of class(you write the name) ,the body of class will be within curly braces {}

```
#include <iostream>
#include <string>
using namespace std;

class Car {

// class body

}
```

Step 2: Create Class Attributes (Properties)

Class Attributes are the variables that store data about an object we use access specifier "public"

```
#include <iostream>
#include <string>
using namespace std;
class Car {
public:
   // These are ATTRIBUTES (Properties)
   string color;    // Stores the car's color
   double price;    // Stores the car's price
   int km;          // Stores the kilometers driven
   string model;    // Stores the car model name
};
```

Step 3: Create Class Methods (Behaviors)

They represent the actions or behaviors of the object.

```
// These are METHODS (Behaviors)
    void start() {
        cout << model << " is starting..." << endl;
    }

    void stop() {
        cout << model << " is stopping..." << endl;
    }

    void forward() {
        cout << model << " is moving forward" << endl;
    }

    void backward() {
        cout << model << " is moving backward" << endl;
    }
};
```

Step 4: Create Objects (Instances)

It's like creating a specific car from the car blueprint. Objects are created inside the main() function

```
int main() {
   // CREATE OBJECT (Instance of Car class)
   Car myCar;

   return 0;
}
```

Step 5: Set Property Values for the Object

Setting Property Values means assigning actual data to the attributes of an object or Reading data from user OR input during initialization

```
int main() {
   Car myCar;

   // SET PROPERTY VALUES
   myCar.color = "Red";
   myCar.price = 23000;
   myCar.km = 1200;
   myCar.model = "Audi";

   return 0;
}
```

Step 6: Call Methods

Calling Methods means executing the functions that belong to an object. It's telling the object to perform its specific actions or behaviors.

```
// Call methods
myCar.start();
myCar.forward();
myCar.backward();
myCar.stop();
```

Complete program

```
#include <iostream>
#include <string>
using namespace std;
class Car {
public:
    // These are ATTRIBUTES (Properties)
    string color;     // Stores the car's color
    double price;     // Stores the car's price
    int km;           // Stores the kilometers driven
    string model;     // Stores the car model name

// These are METHODS (Behaviors)
    void start() {
        cout << model << " is starting..." << endl;
    }

    void stop() {
        cout << model << " is stopping..." << endl;
    }

    void forward() {
        cout << model << " is moving forward" << endl;
    }

    void backward() {
        cout << model << " is moving backward" << endl;
    }
};

int main() {
    Car myCar;

    myCar.color = "Red";
    myCar.price = 23000;
    myCar.km = 1200;
    myCar.model = "Audi";

    // Call methods
    myCar.start();
    myCar.forward();
    myCar.backward();
    myCar.stop();

    return 0;
}
```

Output

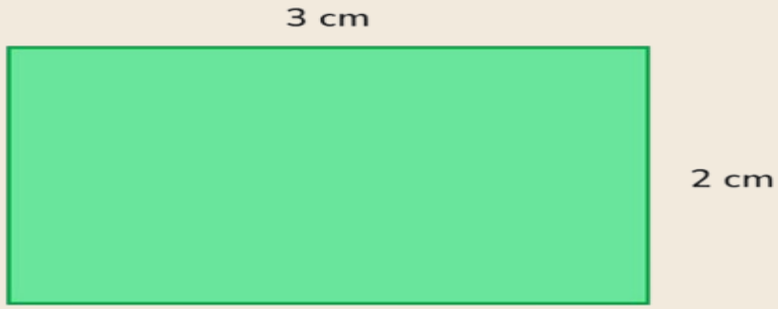
```
Color: red
Price: 23000
KM: 1200
Model: Audi

The red Audi is starting...
The red Audi is moving forward.
The red Audi is moving backward.
The red Audi is stopping.
```

Identifying Attributes and Methods

How to Find Attributes and Methods?

Example : Calacute the area of following rectangle using OOP style?
Area =length * width



🟦 ATTRIBUTES = What Objects HAVE
Nouns like length, width, color
Data storage in classes

🟩 METHODS = What Objects CAN DO
Verbs like calculate calculateArea ()
Result output Display